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IT DEPARTMENT

INTERNSHIP REPORT

OF PERFECTION

T° control system and automatic activation of solenoid valve

Host organization: FUBA Printed Circuits

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Thanks

This report was produced as part of the engineering internship within the company FUBA printed circuits.

First of all, I would like to express my sincere thanks to my supervisor Mr. Jaouhar OUESLATI, maintenance engineer, who helped me enormously and who allowed me to benefit greatly from his experience and his career in this field. He helped me with his advice and recommendations which are effective in developing a reliable and well-structured subject. Also, I would like to thank all the staff, who provided me with the necessary information and knowledge that I needed during my internship. I would also like to thank the respectable members of the jury for agreeing to give me their precious time to comment, discuss and judge my work.

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Chapter 1: Presentation of the host company

1. Introduction:

To find a good result for your project, it is necessary to use successful resources on a large scale.

Here is what I will present in this chapter:

a. Equipment and software used.
b. The code and the assembly of the circuit carried out.

2. Hardware:

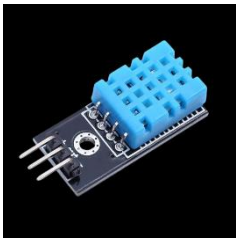
Computer :

- Windows version : Windows 10 Professional 64-bit (Build 17763).
- CPU : Intel(R) Core(TM) i5-6200U.
- RAM : DDR3 8GB.
- Hard drive: 1To.

The programmable card:

The ESP8266 is a microcontroller integrated circuit with Wi-Fi connection developed by the Chinese manufacturer Espressif. The ESP8266 NodeMcu has 16 GPIO pins and one analog input pin, however, only 10 of these GPIO pins can be used for digital input and output operations.

Temperature sensor:



The DHT11 is a very low cost basic digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure ambient air and spits out a digital signal on the data pin.

Relay :



The relay is a controllable switch which makes it possible to isolate the control part from the power part. It is a component widely used when we want to make the link between electronics and electricity. It is used to open or close a contactor on a power circuit according to a signal between 0 and 5V.

Solenoid valve :



A solenoid valve or solenoid valve is an electrically controlled valve. Thanks to this device, it is possible to act on the flow rate of a fluid in a circuit by an electrical signal.

3. Softwares and Apps:

ARDUINO (IDE):

The Arduino Integrated Development Environment or Arduino Software (IDE) contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions, and a series of menus. It connects to Arduino hardware to download programs and communicate with them.

BLYNK APP:

Blynk is a platform for the Internet of Things (IoT). It allows in particular to design a mobile application (Android and iOS) to control and visualize the data of an embedded system via a public or private cloud server.

The design of the mobile application (Android and iOS) based on widgets (graphic elements) is carried out by simple drag & drop without writing a line of code.

GOOGLE SHEETS:



It is an online spreadsheet that allows, among other things, to design spreadsheets, format them and share them with other users via Google Drive. To work, Google Sheets requires nothing more than a connection and a web browser.

4. Connecting the card to the APP :

Here I used the authentication code given by Blynk, FUBA's network ssid and password also the port of course to set up the connection between Blynk and the esp8266.

```
//setting up blynk and linking it to the code
Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
//----------------------------
```

5. Linking google sheets to the card :

Here I'm connecting the card to the client who's going to send the sensor values to google sheets.

```
void update_google_sheet()
{
    Serial.print("connecting to ");
    Serial.println(host);


    WiFiClientSecure client;
    const int httpPort = 443;

    client.setInsecure();

    if (!client.connect(host, httpPort)) {
      Serial.println("connection failed");
      return;
    }


    //----------------------------------------Processing data and sending data
    String url = "/macros/s/" + GAS_ID + "/exec?temperature=";

    url += String(Temperature);

    url += "&humidity=";
    url += String(Humidity);

    Serial.print("Requesting URL: ");
    Serial.println(url);

    // This will send the request to the server
    client.print(String("GET ") + url + " HTTP/1.1\r\n" +
                 "Host: " + host + "\r\n" +
                 "Connection: close\r\n\r\n");

    Serial.println();
    Serial.println("closing connection");
}
```

6. Sending data to google sheets:

As you can see here the values are being sent to the client and doing an update according to an interval of time (specific period) every 5 minutes.

```
void sheet(){

  time_ms = millis();
  time_dif = time_ms - time_1000_ms_buf;

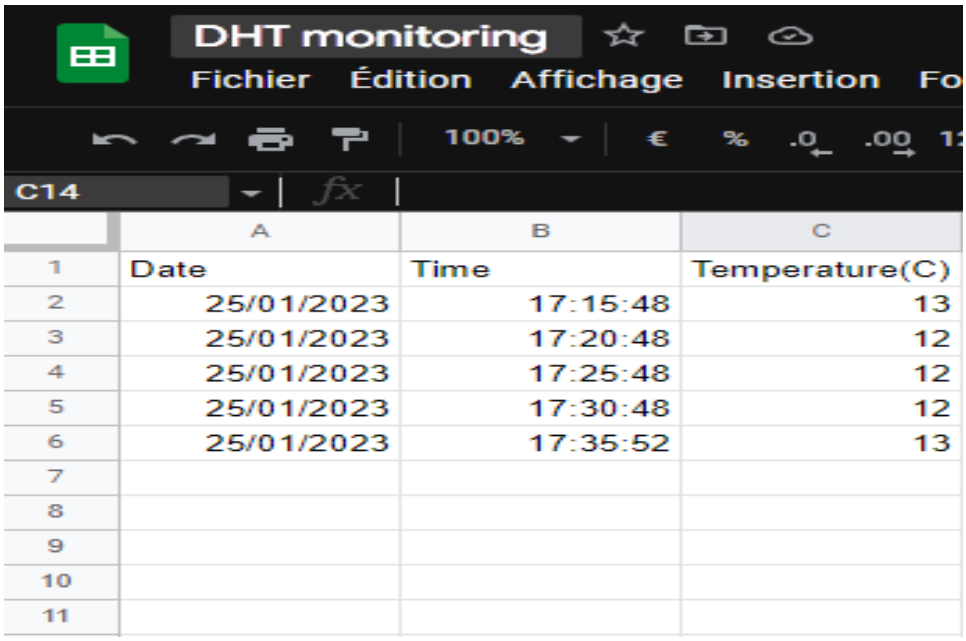
  // Read and print serial data every 1 sec
  if ( time_dif >= 1000 ) // 1sec
  {
    time_1000_ms_buf = time_ms;
    Temperature = dht.readTemperature();
    Humidity = dht.readHumidity();

    //Print serial messages
    Serial.print("Humidity: " + String(Humidity) + " %");
    Serial.print("\t");
    Serial.println("Temperature: " + String(Temperature) + " C");

    digitalWrite(LED_BUILTIN, !digitalRead(LED_BUILTIN));
  }

  // Update data to google sheet in specific period
  time_ms = millis();
  time_dif = time_ms - time_sheet_update_buf;
  if ( time_dif >= UPDATE_INTERVAL_MS ) // Specific period
  {
    time_sheet_update_buf = time_ms;
    update_google_sheet();
  }

}
```



	A	B	C
1	Date	Time	Temperature(C)
2	25/01/2023	17:15:48	13
3	25/01/2023	17:20:48	12
4	25/01/2023	17:25:48	12
5	25/01/2023	17:30:48	12
6	25/01/2023	17:35:52	13
7			
8			
9			
10			
11			

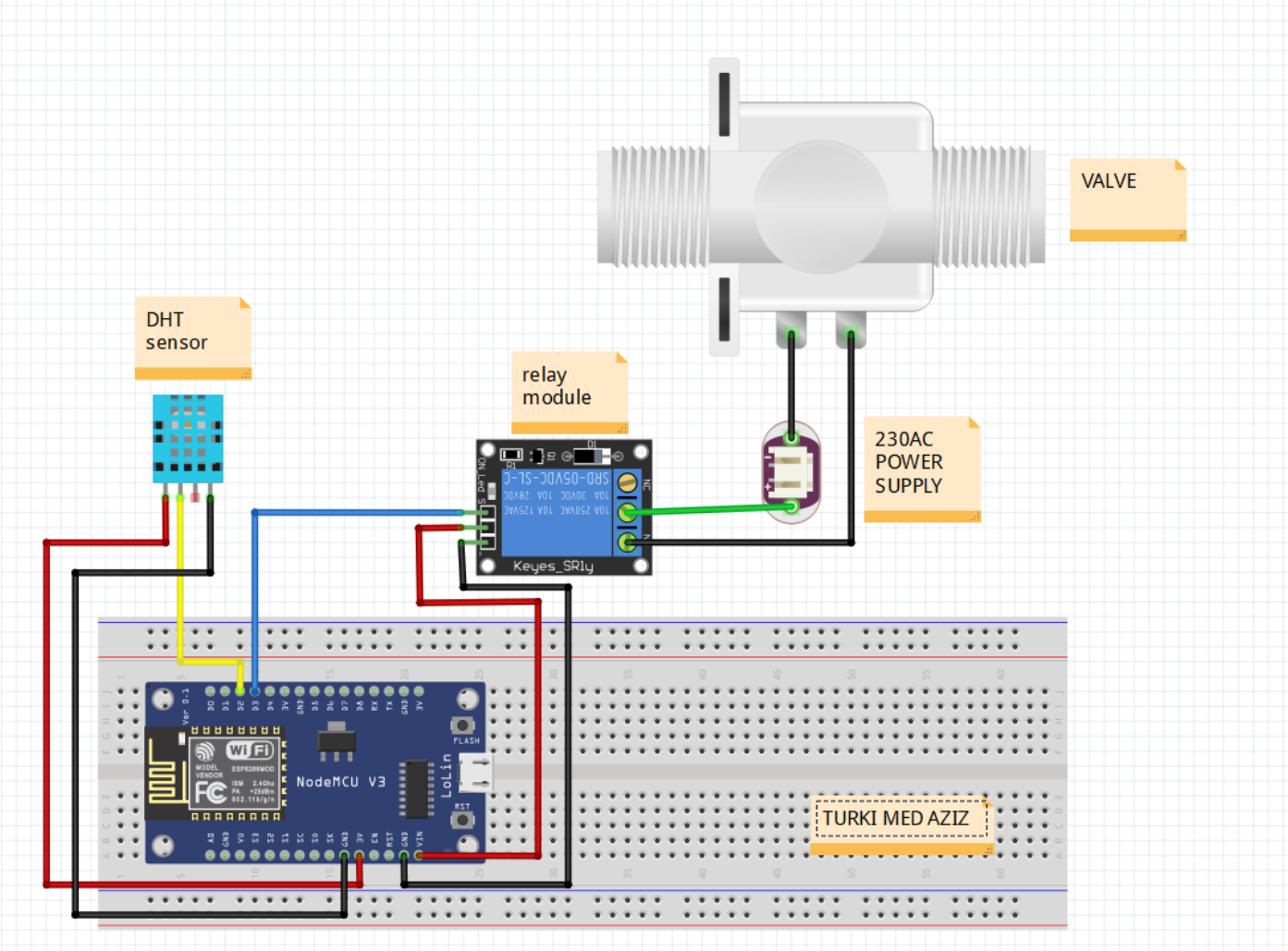
7. Reading data on the APP:

Here's the app Blynk. This interface shows the gauge of both temperature and humidity, with a slide button (ON/OFF) which turns the solenoid valve on or off manually.



8. Circuit:

Here's the circuit with the main components.



9. Conclusion:

In this chapter, I presented the material, software; environments, the main functions and a circuit.

General Conclusion

This report is the summary of a professional project carried out at FUBA Company which lasted a month in which I tried to present the main functions to make the google sheets and the sensor work together. This project is specially developed for the use of the security manager.

This internship perfectly met my expectations because I wanted to discover the field of Embedded System in a professional life, have a real experience not only based on theories. It allowed me to discover a universe that I ultimately knew very little about it in which I have a huge interest.

For mainly sure this project can be upgraded for the best and I intend to work on it, add some more features and functions.

Finally, I hope that my work would be up to the task in wishing that. I reach the target of this report